

Ziad Alzarka

Senior Software Engineer | Platform, Product & AI

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SUMMARY

Senior software engineer and systems architect with **10 years** building **data-intensive platforms** across **finance and healthcare**: ledgers, reconciliation and AP automation; AI medical assistants, claims and lab pipelines. Takes products from architecture to production across backend, data, web and mobile, and builds **AI agents and workflows that automate engineering**.

WORK EXPERIENCE

Light  | Copenhagen, Denmark (Remote)

Dec 2025 - Present

Senior Software Engineer, Procure-to-Pay & Record-to-Report

Worked on the ledger's reliability and the finance products built on it, from customer conversations to production.

- Led **intercompany invoicing** end to end, from domain research and design to building every layer.
- Built **three-way matching** with customers who needed it to move their AP onto the platform: automatic bill-to-purchase-order matching, and goods-receipt confirmation built across every layer in two weeks.
- Root-caused and fixed **three production incidents**: a platform outage, connection-pool exhaustion and out-of-memory pod crashes; added platform-wide safeguards against repeats.
- Sped up ledger queries and exports: reclaimed **~17.5 GB** of duplicate and unused indexes, added indexes for real query patterns, moved heavy reads to a replica and made large exports resumable background jobs.
- Cut failed bank payments from **~3% to near zero** with automatic retries, and made the integration test suite **~10x faster** with per-test data isolation and 10 parallel partitions.
- Designed a reusable **notification system**: new notification types and delivery channels such as Slack and Microsoft Teams could plug in without changing the core, with async delivery, retries and do-not-disturb.
- Set the team's **AI tooling standards**. Built an AI code reviewer that reviewed **1,700+ PRs** and prevented **45+ bugs**, and a security agent that prevented **5+ vulnerabilities**.
- Built **Lucid**, the repo's docs and ADR system, to spread knowledge beyond a few engineers, which helped new engineers onboard faster.
- Made **Datadog metrics and alerts** the team's default in every PR.
- Delivered **aged AP/AR reports, EU Sales List VAT reporting and non-calendar fiscal years**, replacing manual year-end SQL work and clearing statutory reporting blockers for customer go-lives.

Ankor.app | Malmö, Sweden (Remote)

Jul 2025 - Nov 2025

Senior Software Engineer

Led technical development of a financial reconciliation platform for e-commerce businesses, from inception to production.

- Architected a multi-tenant platform processing **millions of transactions monthly at 98%+ reconciliation accuracy**, integrating 5+ payment providers (Shopify, Klarna, PayPal, Plaid, Adyen).
- Built a four-tier dbt and Python pipeline (raw → staging → fact → app) over PostgreSQL, BigQuery and MotherDuck for transaction matching, multi-currency support and automated double-entry journals.
- Delivered the full product across a React/TypeScript frontend, GraphQL API and NestJS backend, with monorepo tooling and CI/CD to GCP.

Senior Software Engineer

Led technical development of an AI medical assistant for laboratories, physicians and insurers.

- Built **AI-assisted diagnosis and test recommendations**, plus insurance claim filing and fee schedules.
- **Recovered a completely lost production database of ~1.5 billion records** by replaying events reconstructed from Kafka, CloudWatch and S3.
- Migrated the primary datastore from **Cassandra to DynamoDB**, improving availability and performance while cutting maintenance and cost by **90%**.
- Introduced a graph database for relationship-heavy queries, **cutting aggregation latency by 35%**; removed race conditions and ACL overhead in data services, **cutting latency a further 20%**.
- Integrated **vector search into the medical knowledge graph** on AWS OpenSearch, built ingestion pipelines for unstructured medical histories, and wrote software streaming results from Sysmex lab analysers.
- Built real-time cross-database replication and set the team's CI, code-standards and RFC practices.

PROJECTS

Kin.app [🔗](#) | [App Store](#) · [Google Play](#) · [Website](#)

Jan 2026 - Present

Designed, built and shipped solo: an AI-powered app for splitting receipts by item.

- Built an **AI receipt reader** (~0.8¢, ~9s per scan) that escalates to a stronger model when OCR and a small model disagree, reaching **97% accuracy** on 500 receipts across scripts, languages, tax rules and formats.
- Built a **voice assistant** that turns live speech into receipts, and an **MCP server** for personal AI agents.
- Built a **double-entry ledger**, and **150+ fitness functions** adapted from *Software Architecture: The Hard Parts* that govern how AI changes the codebase and keep quality consistent.

peel [🔗](#) | [GitHub](#)

Jul 2026 - Present

Open-source Go TUI that combines code review and Git staging, with **Claude Code and Codex walkthroughs**.

- Built comments that follow code through edits, with structured feedback for coding agents.

iCloud Image Labeler [🔗](#) | [GitHub](#)

Python tool that makes Apple Photos searchable with AI keywords, captions and OCR, using **local models**.

EARLIER EXPERIENCE

CoatConnect | Software Engineer, Cairo, Egypt

2019 - 2020

Developed a medical events platform, refactored it end to end and optimized its performance to **1.7s LCP** ([article](#)).

BirdCloud | Software Engineer, Cairo, Egypt

2017 - 2018

Accelerated product delivery with a reusable component library, plus reusable ERP modules and a C# library, increasing client capacity by **20%** and revenue by **30%**.

Magmooty | formerly Coretrix, Cairo, Egypt

2016 - Present

Built a platform for Egyptian tutoring centres, later rebuilt as an offline-first Rust app with Go-to-Rust FFI ([article](#)).

SKILLS

Languages: Kotlin, TypeScript, Python, Rust, Go, SQL, C#, Java

Data: PostgreSQL, Redis, Elasticsearch, DynamoDB, MongoDB, DocumentDB, AWS Neptune, Neo4j, vector search

Platforms: AWS, GCP, Docker, Kubernetes, Terraform, Datadog, Grafana, CI/CD, React, React Native

AI: Claude Code, Codex, MCP, AI tuning, LLM evaluation

EDUCATION

Bachelor of Science, Computer Science

2018 - 2022

English: bilingual · Arabic: native